

第2章 Data 資料源

- R/Python/Julia/SQL程式設計與應用
(R/Python/Julia/SQL Programming and Application)
- 資料視覺化 (Data Visualization)
- 機器學習 (Machine Learning)
- 統計品管 (Statistical Quality Control)
- 最佳化 (Optimization)



李明昌博士

<https://www.youtube.com/@alan9956>

<http://rwepa.blogspot.com/>

alan9956@gmail.com



大綱

1. 認識Data介面

1.1 加入Data

1.2 欄位設定

1.3 Type資料類型

2. Expression Assistant (運算式小幫手)

Excel公式 **=TODAY ()**

3. Virtual Column (虛擬欄位)

1.認識Data介面

- 1.1 加入Data
- 1.2 欄位設定
- 1.3 Type資料類型



1.1 加入Data

(Data 表示工作表)

自動新增第1個資料表的Data-員工清單

- 參考第1章-建立應用程式: appsheet_demo

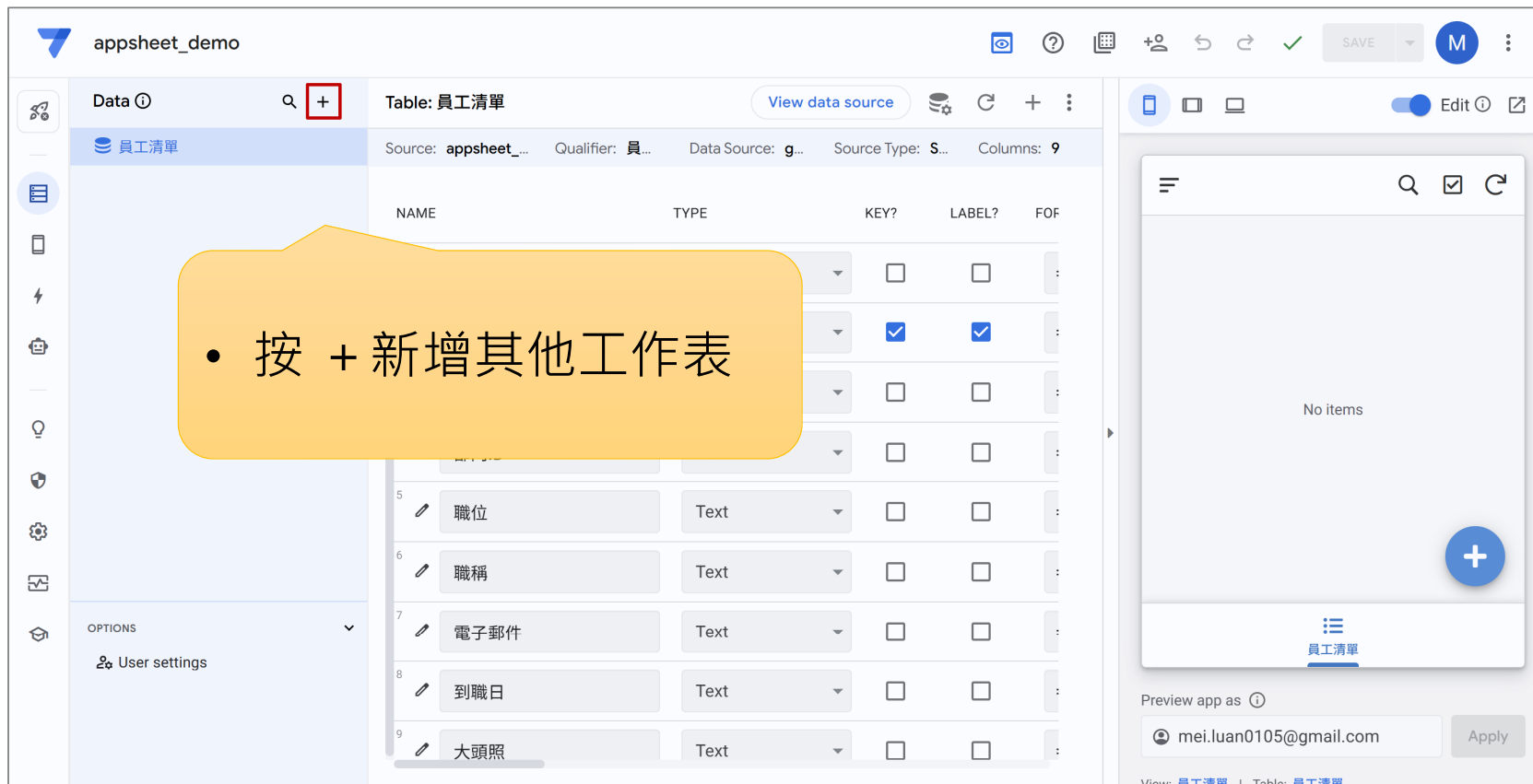


Table: 員工清單

Source: appsheet_... Qualifier: 員... Data Source: g... Source Type: S... Columns: 9

	NAME	TYPE	KEY?	LABEL?	FOF
1			☐	☐	
2			☒	☒	
3			☐	☐	
4			☐	☐	
5	職位	Text	☐	☐	
6	職稱	Text	☐	☐	
7	電子郵件	Text	☐	☐	
8	到職日	Text	☐	☐	
9	大頭照	Text	☐	☐	

按 + 新增其他工作表

Preview app as mei.luan0105@gmail.com Apply

新增 “部門清單” Data

Add data

SUGGESTIONS: ADD MORE DATA FROM CONNECTED SOURCES

Add Table "部門清單"

Add Table "請假紀錄"

Add Table

Add Table "薪資清單"

Add Table "引導頁面"

← Create a new table

Source Id
A provider-specific path to the table data (not user-visible)

appsheet_demo

Worksheet Name/Qualifier
Further information used to access this data (e.g. the worksheet name for a spreadsheet data source)

部門清單

Are updates allowed?
Are adds, deletes and updates allowed?

✓ Updates

✓ Adds

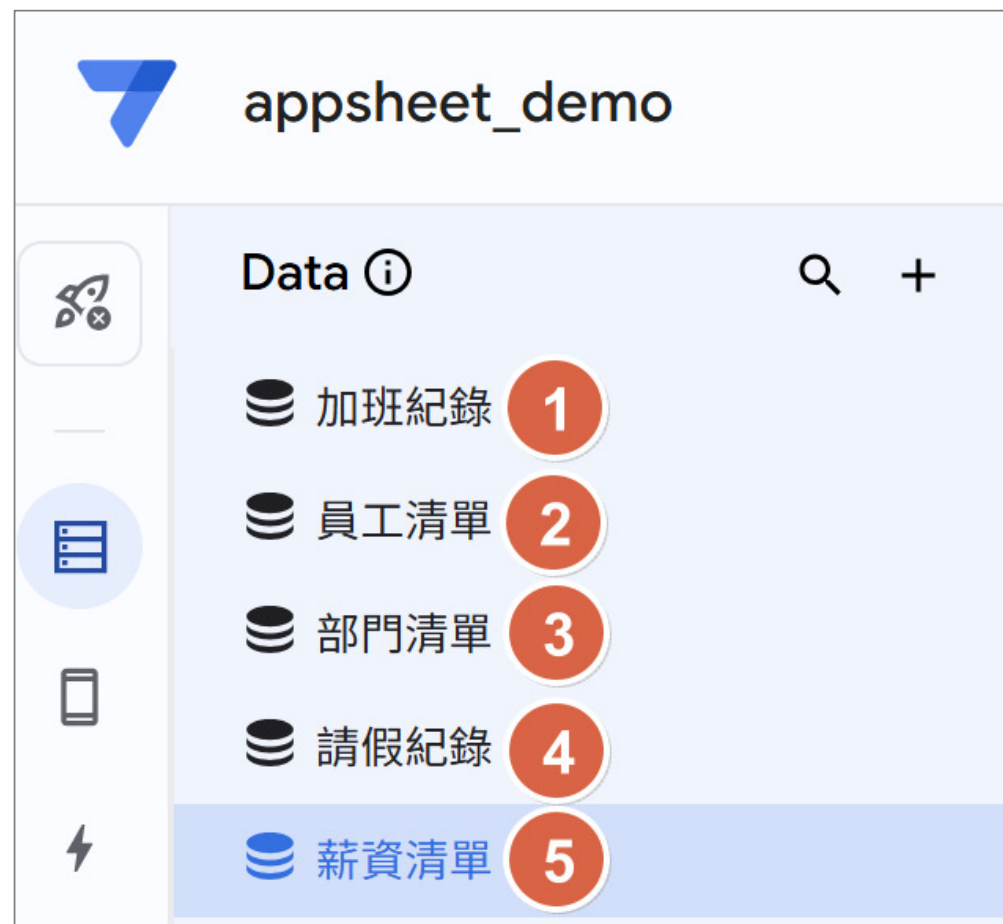
✓ Deletes

Read-Only

Add to app



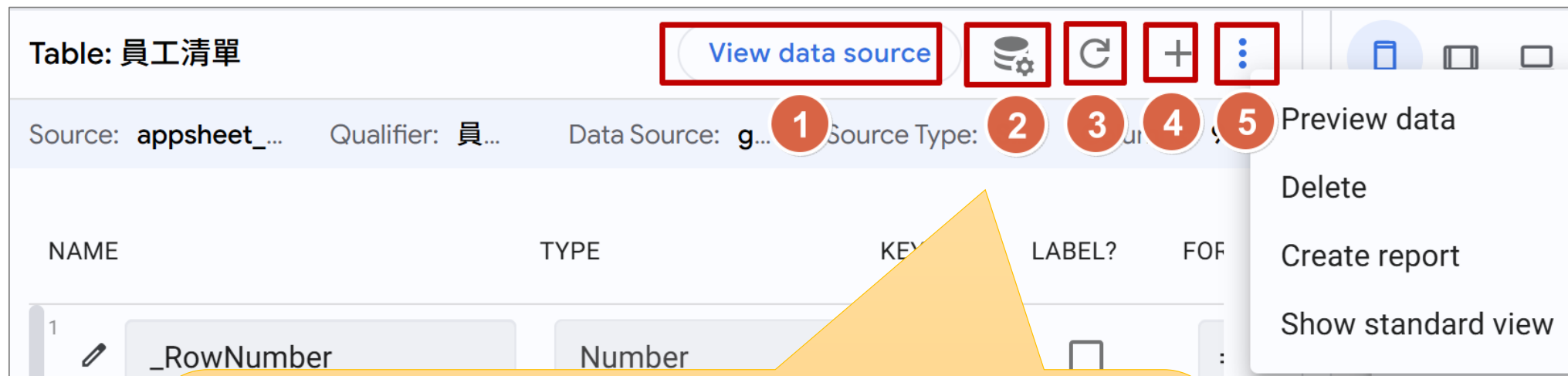
新增5個 Data – 完成版





1.2 欄位設定

Data – 工具列



The screenshot shows the AppSheet interface for a table named "Table: 員工清單". The toolbar at the top contains several icons: "View data source" (highlighted with a red box), a database icon (highlighted with a red box), a refresh icon (highlighted with a red box), a plus icon (highlighted with a red box), and a more options icon (highlighted with a red box). Below the toolbar, the table's source information is displayed: "Source: appsheet_...", "Qualifier: 員...", "Data Source: g...", "Source Type: ", and "Preview data". The table structure is shown below, with columns: NAME, TYPE, KEY, LABEL?, and FOF. The first row is highlighted, showing "_RowNumber" and "Number". A yellow callout box points to the toolbar icons, listing their functions.

NAME	TYPE	KEY	LABEL?	FOF
1	_RowNumber	Number		

1. View data source: 檢視資料來源，開啟Google試算表
2. Table settings: 表格設定
3. Regenerate schema: 重新載入表格，適用資料新增，變更
4. Add virtual column: 新增 Virtual Column (虛擬欄位)
5. More: 進階設定

Table settings: 表格設定

Table settings

Table name

Name for this table (user-visible)

員工清單

Are updates allowed?

Are adds, deletes and updates allowed?

✓ Updates

✓ Adds

✓ Deletes

Read-Only



Storage

Source Path

A user-understandable path to the table data

appsheet_demo

Worksheet Name/Qualifier

Further information used to access this data (e.g: the worksheet name for a spreadsheet data source)

員工清單

Data Source

Name of the table data source

google

Copy Data to New Source

Source Type

A source type for the specific provider (not user-visible)

 Google Sheets

Done

一般不用更改

@RWEPA

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欄位設定

appsheet_demo

Table: 員工清單

Source: appsheet_demo Qualifier: 員工清單 Data Source: google Source Type: Sheets Columns: 9

欄位名稱 NAME

• 不可重複

• 不可使用特殊符號

資料類型 TYPE

是否為主鍵 KEY?

原始表格8個欄位, 自動新增 _RowNumber

合計: 9個欄位

主鍵特性:

1. 可以識別一筆記錄

2. 不可重複

3. 不可以空白

展開右側窗格

NAME	TYPE	KEY	LABEL?	FORMULA	SHOW?
_RowNumber	Number	☐	☐	=	☐
		☐	☐	=	☒
		☐	☐	=	☒
	Text	☐	☐	=	☒
	Text	☐	☐	=	☒
6 職稱					
7 電子郵件					
8 到職日					
9 大頭照	Text	☐	☐	=	☒

欄位設定

- EDITABLE? : 是否可以編輯
- REQUIRE? : 是否為必填資料
- INITIAL VALUE : 預設值
- DISPLAY NAME : 顯示的名稱, 不一定要與 NAME 相同
例: NAME為英文字，但畫面顯示中文字型
- DESCRIPTION : 描述欄位的用途

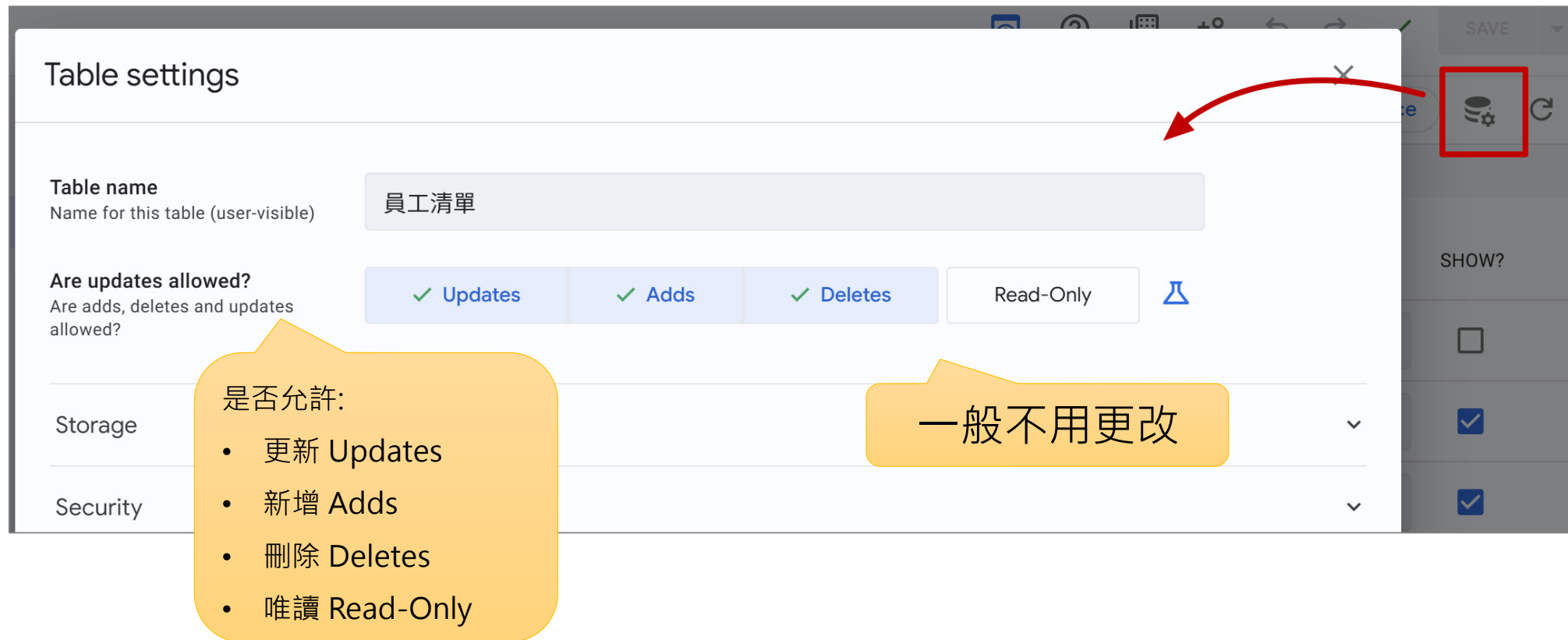
EDITABLE?	REQUIRE?	INITIAL VALUE	DISPLAY NAME	DESCRIPTION
☐	☐	=	=	= Number of this row
☒	☒	= UNIQUEID()	=	=
☒	☐	=	=	=

欄位設定 (續)

- SEARCH? : 此欄位是否可以使用搜尋
- SCAN? : 支援條碼掃描
- NFC : 支援近場通訊 (Near-Field Communication)
- PII : 個人識別資訊 (Personally Identifiable Information)

SEARCH?	SCAN?	NFC?	PII?
☐	☐	☐	☐
☒	☐	☐	☐
☒	☐	☐	☐

Table setting 表格設定



The screenshot shows the 'Table settings' dialog in AppSheet. A red box highlights the settings icon in the top right corner of the app interface, with a red arrow pointing to the 'Table settings' dialog. The dialog has a title bar with a close button (X). The settings are as follows:

- Table name:** 員工清單 (Employee List). The label below it says 'Name for this table (user-visible)'.
- Are updates allowed?** Are adds, deletes and updates allowed?
 - ☒ Updates
 - ☒ Adds
 - ☒ Deletes
 - ☐ Read-Only
- Storage:** (Dropdown menu)
- Security:** (Dropdown menu)

Two yellow callout boxes provide additional information:

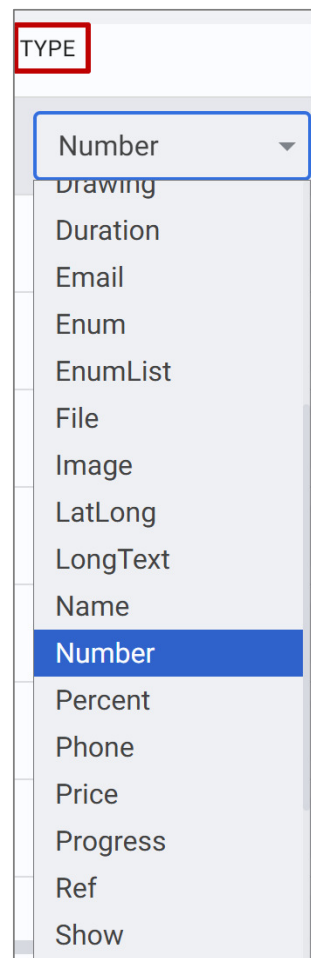
- A callout box on the left, titled '是否允許:' (Whether allowed:), lists the options: 更新 Updates, 新增 Adds, 刪除 Deletes, and 唯讀 Read-Only.
- A callout box on the right, titled '一般不用更改' (Generally no need to change), points to the 'Read-Only' option.



1.3 Type資料類型

資料類型 vs. Excel 數值格式

AppSheet: Type



Excel: 數值格式



資料類型 (續)

- <https://support.google.com/appsheet/answer/10106435>

Column data type	說明	Types
Communication types	通訊類型	Email, Phone
Content types	內容類型	Drawing, File, Image, Signature, Thumbnail, Video
Enumerated types	枚舉類型	Color, Enum (單選) , EnumList (複選) , List, Progress, Ref, Yes/No
Mappable types	可映射類型	Address, LatLong, XY
Numeric types	數字類型	Decimal, Number, Note, Percent, Price
Show types	顯示類型	Show
Temporal types	時間類型	Date, DateTime, Duration, Time
Text types	文字類型	LongText, Name, Text
Other types	其他類型	URL, App, Multi Column Key, ...



Type: Email (電子郵件)

Type: Email

The screenshot displays the AppSheet configuration interface for a table named 'Table: 員工清單' (Table: Employee List). The left sidebar shows the 'Data' section with various tables, including '員工清單' (Employee List) which is currently selected. The main area shows the configuration for the '員工清單' table, with columns including NAME, TYPE, KEY?, LABEL?, and FOF. The '電子郵件' (Email) column is highlighted with a red box, and its type is set to 'Email'. A red arrow points from this configuration to a preview of the form on the right. In the preview, the '電子郵件' field contains the text 'alan9956', and a red error message below it states 'This entry is invalid'. A yellow speech bubble with the text '自動驗證Email' (Automatically verify Email) points to the error message. The bottom of the preview shows a 'Preview app as' section with the email 'mei.luan0105@gmail.com' and an 'Apply' button.

NAME	TYPE	KEY?	LABEL?	FOF
1	_RowNumber	Number	☐	☐
2	ID	Text	☒	☒
3	員工姓名	Text	☐	☐
4	部門ID	Text	☐	☐
5	職位	Text	☐	☐
6	職稱	Text	☐	☐
7	電子郵件	Email	☐	☐
8	到職日	Text	☐	☐
9	大頭照	Text	☐	☐

Type: EnumList (下拉式清單)

Type: EnumList

5

✎ 職位

EnumList



員工清單：職位

type: EnumList

Done

Type

Column data type

EnumList

Type Details

Values

⋮ 總經理

⋮ 協理

⋮ 經理

⋮ 課長

⋮ 職員

✎

✖

新增

Add

Type: EnumList (續)

員工清單：職位

type: EnumList

Done

✕

Allow other values

☐

允許手動新增輸入

Auto-complete other values

☐

自動匹配現有值

Base type

顯示方式: 自動 / 按鈕 / 垂直排列 / 下拉式選單

Input mode

Auto

Buttons

Stack

Dropdown

Item separator

項目區隔符號

顯示方式: 按鈕 / 垂直排列/ 下拉式選單

職位

總經理

協理

經理

課長

職員

職位

總經理

協理

經理

課長

職員

職位

☒ 總經理

☐ 協理

☐ 經理

☐ 課長

☐ 職員

複選: 正方形方塊

Type: Number (數值)

Type: Number

- `_RowNumber` (列編號)：自動產生之欄位, 一般不用更改.

NAME	TYPE	KEY?	LABEL?	FORMULA	SHOW?
1  <code>_RowNumber</code>	Number ▼	☐	☐	= 	☐ 

Type: Number (續)

員工清單 : _RowNumber
type: Number Done 

Column name
Should match the column name in the data source

[Go to display options](#) ⓘ

Show?
Is this column visible in the app?
You can also provide a 'Show_If' expression to decide.
☐ 

Type
Column data type

Type Details

Numeric digits - +

Show thousands separator ☒

- 小數點位數
- _RowNumber 資料類型至少須1位

顯示千分位符號

Type: Number - Data Validity (資料驗證)

員工清單 : _RowNumber

type: Number

Done

✕

Data Validity

Valid If
Does the column contain valid data?

=



Invalid value error
Error message when the value is not valid.

=



Require?
Are users required to fill in this column before saving a row? You can also provide a 'Required_If' expression to decide.

☐



Type: Number- Auto Compute (自動計算)

員工清單 : _RowNumber

type: Number

Done

⌵

Auto Compute

App formula
Compute the value for this column instead of allowing user input.

=

公式: [數量]*[單價]

⌵

Initial value
The default starting value for this column.

=

初始值: TODAY ()

⌵

Suggested values
An optional expression with a list of suggested values for this column.

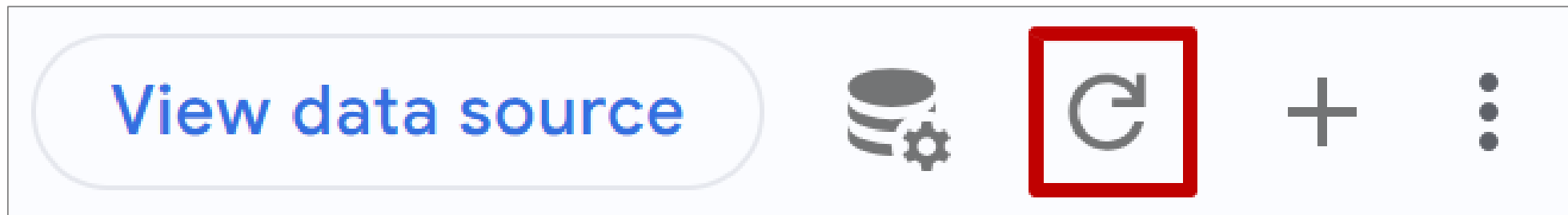
=

建議值: SELECT (城市[City], TRUE)
城市工作表中 City 欄位

⌵

Spreadsheet formula
Generated from your spreadsheet.
Gets applied during sync.

Regenerate schema



Regenerate schema

重新整理綱要 (資料)



函數總表

- AppSheet 函數清單, 全部有124個函數:
https://github.com/rwepa/teaching-appsheet/blob/main/AppSheet_Function_List.pdf
- 參考: <https://support.google.com/appsheet/table/10104782?hl=en>
- 函數類型: 全部12個, Conditional, Date and time, Image, Informational, Link, List, Location, Logical, Math, Table, Text, Yes/No

	A	B	C	D	E
1	No	Type	Name	Syntax	Returns
2	1	Conditional	IF	IF(logical-expression,value-if-true,value-if-false)	Either/or conditional evaluation.
3	2	Conditional	IFS	IFS(condition1,value1, [condition2,value2 ...])	First-match conditional evaluation.
4	3	Conditional	SWITCH	SWITCH(expression,case1,value1, [case2,value2...],default)	Choose-one conditional evaluation.
5	4	Date and time	DATE	DATE(when)	DatefromDate,DateTime, orTime.
6	5	Date and time	DATETIME	DATETIME(when)	DateTimefromDate,DateTime, orTime.

2. Expression Assistant (運算式小幫手)

Expression Assistant

- 運算式小幫手可以使用於 Formula (公式) , INITIAL VALUE (預設值)
- 運算式小幫手沒有區分英文字母大小寫, 建議統一使用大寫
- 範例: Table: 員工清單 → 到職日 → edit → Auto Compute

常用Expression Assistant (運算式小幫手):

<https://github.com/rwepa/teaching-appsheet/blob/main/ExpressionAssistant.pdf>

Expression Assistant 範例 - edit

appsheet_demo

Data ⓘ

- 加班紀錄
- 員工清單** 1
- 部門清單
- 請假紀錄
- 薪資清單

Table: 員工清單

Source: appsh... Qualifier: ... Data Sour... Source Ty... Columns: 9

NAME	TYPE	KEY?
ID	Text	☒
員工姓名	Text	☐
部門ID	Text	☐
職位	Text	☐
職稱	Text	☐
電子郵件	Text	☐
到職日	Text	☐

edit 2

OPTIONS

- User settings

Expression Assistant 範例 - App formula

員工清單：到職日

type: Text

Done

✕

Auto Compute

App formula

Compute the value for this column instead of allowing user input.

=

Initial value

The default starting value for this column.

=

Suggested values

An optional expression with a list of suggested values for this column.

=

Spreadsheet formula

Generated from your spreadsheet. Gets applied during sync.

輸入 TODAY ()

Expression Assistant

App Formula for column 到職日 (Text)

TODAY() 1

綠色打勾表示正確

Test

Examples Data Explorer

Yes/No	Math	Text	Time	Columns	Lists	Deep Links	Other
TODAY()			Date	TODAY()			Insert
TIMENOW()			Time	TIMENOW()			Insert

Help me with expressions

Cancel Save 2

3.Virtual Column (虛擬欄位)

Virtual Column (虛擬欄位)

- Virtual Column (虛擬欄位) 不實際儲存在資料庫 (例如: Google Sheet) 中的欄位。
- 虛擬欄位的值是使用運算式計算產生，使用者不可自行輸入。
- 虛擬欄位的值是具有即時計算的特性，每次資料變動時將重新計算。

特點	說明
不寫入資料表	只存在 AppSheet，不會新增資料列到 Google Sheet
自動更新	每次同步或資料變更就會重新計算
用於關聯、彙整、計算	用於 Ref、Related 列表、自動計算的欄位
影響效能	過多 Virtual Columns 會降低效能

虛擬欄位使用範例

- 計算欄位 (類似 Excel)
- 聚合資料 (SUM、COUNT、LIST)
- 建立 Related Rows (自動產生子報表)
- 顯示計算後的文字、日期、狀態
- 格式化使用 (顯示資訊於畫面)

Add virtual column

appsheets_demo

Data ⓘ

Table: 加班紀錄

Source: appsh... Qualifier: ... Data Sourc... Source Ty... Columns:

加班紀錄

員工清單

部門清單

請假紀錄

薪資清單

OPTIONS

User settings

NAME	TYPE	KEY?
加班日期	Text	☐
加班起始時間	Text	☐
加班結束時間	Text	☐
加班事由	Text	☐
加班時數	Text	☐
主管員工ID	Text	☐
審核狀態	Text	☐
最後更新時間	Text	☐

View data source

Preview data

Delete

Table settings

Regenerate schema

Add virtual column

Create report

Show standard view

員工清單

加班起始日期與時間

- Column name: 加班起始日期與時間

加班紀錄 : 加班起始日期與時間 (virtual)
type: Unknown

Delete

×

Column name
Column name

加班起始日期與時間 1

App formula
Compute the value for this column instead of allowing user input.

= 2

⚙

Expression Assistant

- `CONCATENATE(TEXT([加班日期]), " ", TEXT([加班起始時間]))`

Expression Assistant

App Formula for column New Virtual Column

`CONCATENATE(TEXT([加班日期]), " ", TEXT([加班起始時間]))`



Test 

加班起始日期與時間 – 完成版

Expression Assistant

App Formula for column 加班起始日期與時間

1

CONCATENATE(TEXT([加班日期]), " ", TEXT([加班起始時間]))

1

✓

Test [↗](#)

Examples

Data Explorer

Yes/No

Math

Text

Time

Columns

Lists

Deep Links

Other

Pattern	Result	Example	
[_THIS]	Unknown	[_THIS]	Insert
[{column_name}]	Number	[_RowNumber]	Insert 2

[Help me with expressions](#)

Cancel

Save

謝謝您的聆聽

Q & A

李明昌

alan9956@gmail.com

<https://www.youtube.com/@alan9956>

<http://rwepa.blogspot.tw/>

